

Unnamed_Unit

Unit 2 - Bringing Ideas to Life

PDF Documents

PDF Document 1: 1738070296214-Chapter 7 - Unit 2 - Bringing Ideas to Life.pdf

This PDF document is included in the unit materials.



UNIT 2 BRINGING IDEAS TO LIFE

Project Introduction

FINDING FAIRNESS: THE CAKE CONTEST



In the bustling classroom of Lux Middle School, a friendly dispute arose between two eighth-graders, Hugh and Cathy, over a delicious batch of cakes. Each felt deserving of a larger share, sparking a lively debate that caught the attention of their teacher, Ashton.

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Seeking a fair solution, Ashton proposed a playful competition: whoever won a series of games would claim all the cakes. Hugh and Cathy agreed, eager to prove their gaming prowess.

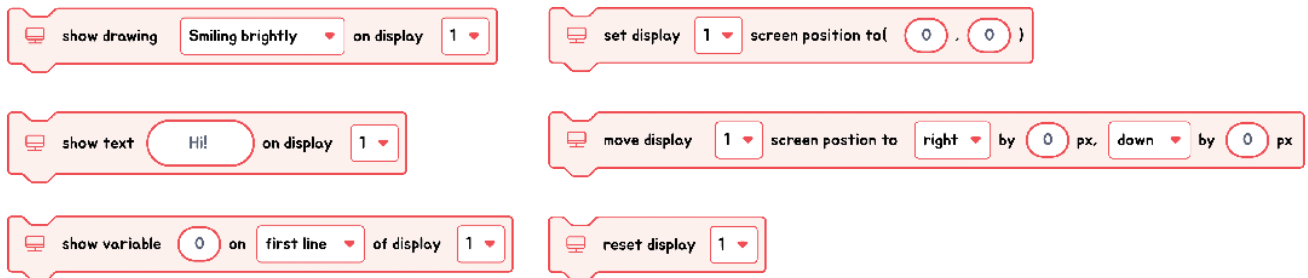
As the games progressed, Cathy consistently emerged victorious, much to Hugh's dismay. Feeling disheartened, Hugh questioned the fairness of the games, delaying the resolution.



Recognizing the need for impartiality, Ashton suggested a final game—a random dice game. Lacking traditional dice, Hugh and Cathy improvised, using MODI modules to craft their own. With a roll of the improvised dice, Hugh triumphed, claiming all the cakes. However, in a surprising turn of events, he generously offered to share his winnings with Cathy.

In the end, the contest not only settled the dispute but also strengthened the bond between Hugh and Cathy, reminding them that fairness and friendship go hand in hand.

ABOUT DISPLAY MODULE

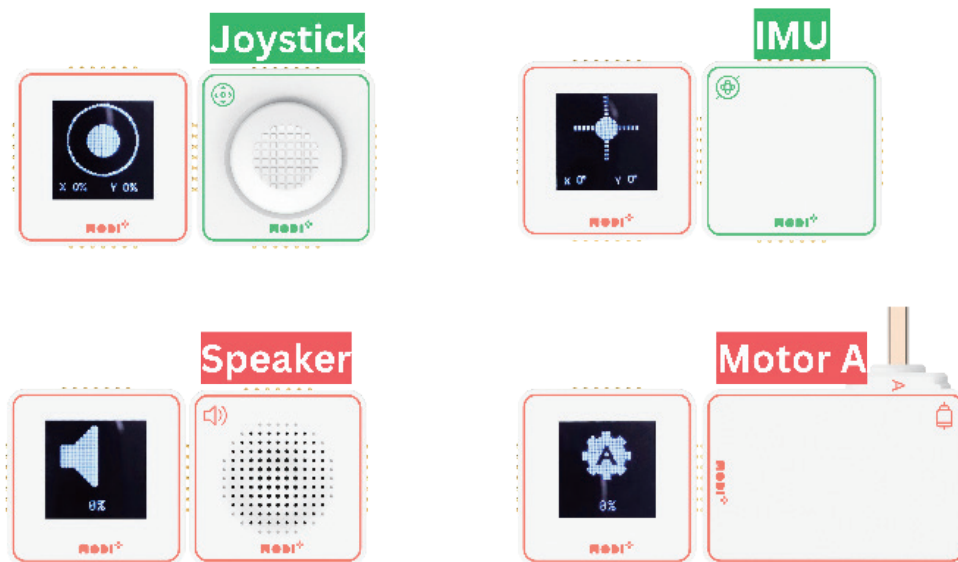


The Display module is like a tiny screen for your MODI projects. It uses 6 different blocks in the Code Editor to show pictures or information from other MODI modules nearby. So, when other modules talk to the Display, it can show you what they're saying or doing with images or data right on its screen!

AUTOMATIC DATA DISPLAY WITH MODI MODULES

The Display module is like the heart of a conversation among MODI modules. Imagine it as a screen that lights up with information whenever other modules, like the Button or the Light Sensor, have something to "say." When any input or output module is connected to it, the Display module automatically picks up their signals and shows what they're doing on its screen.

INTERACTION BETWEEN DISPLAY AND OTHER MODULES



So, if you press a Button module or cover a Light Sensor, the Display will show these actions without needing any special programming. It's like having a translator that instantly shows you what's happening in the MODI module language!



Required Modules & Brainstorming

Based on the story, which MODI modules do you need for this creation?



Button



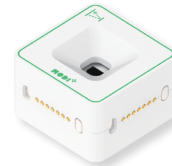
Dial



Environment



Joystick



ToF



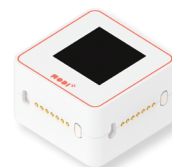
Speaker



IMU



LED



Display

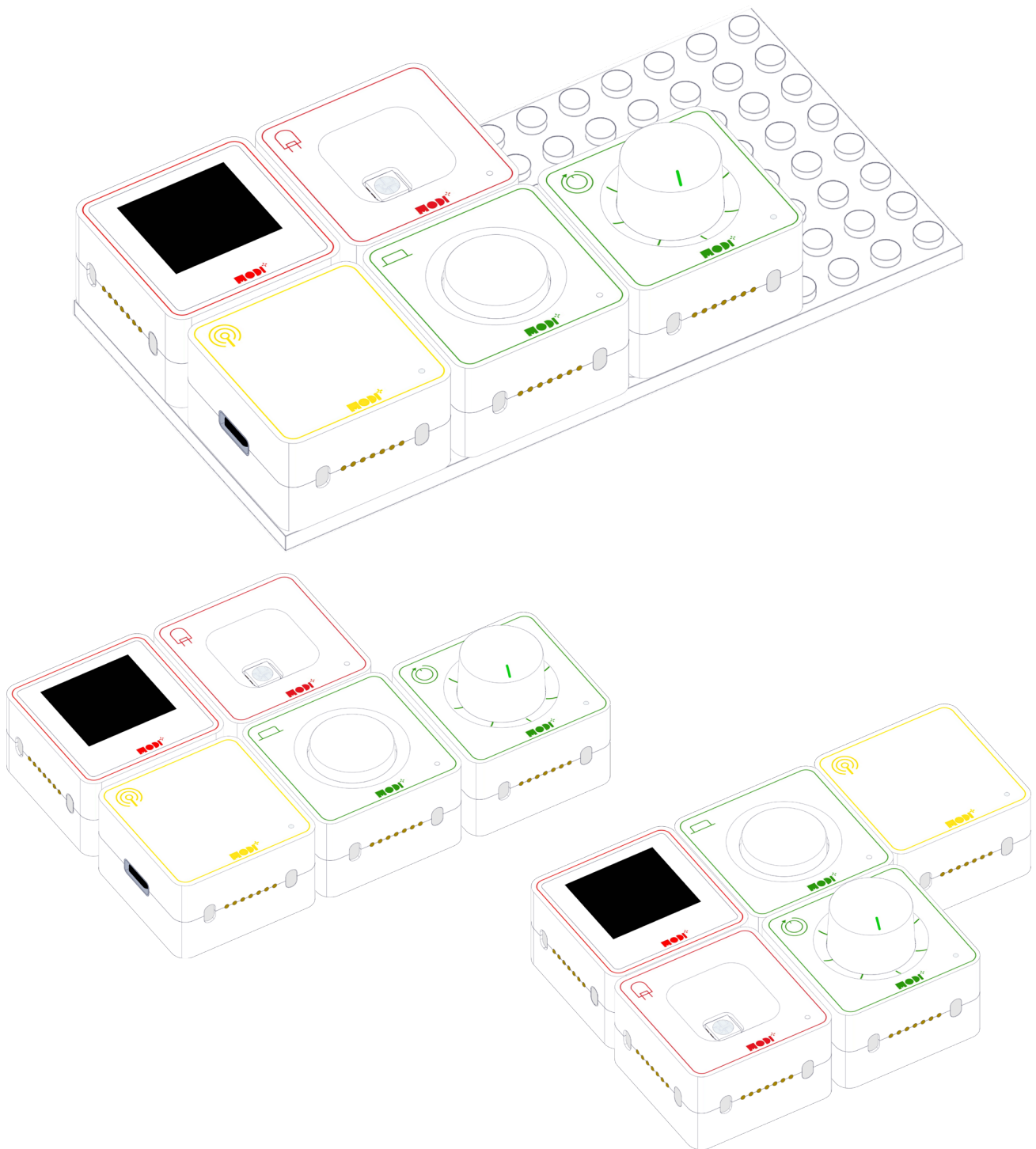


Motor A



Motor B

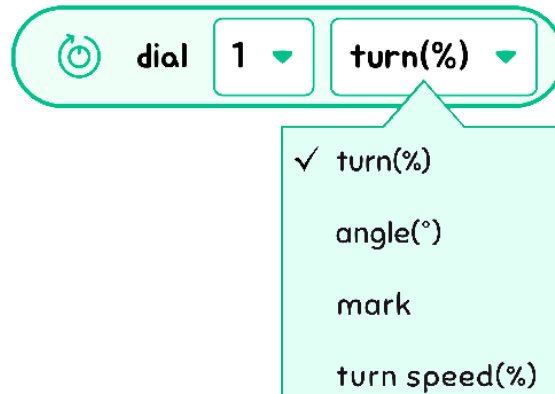
HOW DOES THE CREATION LOOK LIKE?





Module Functionality

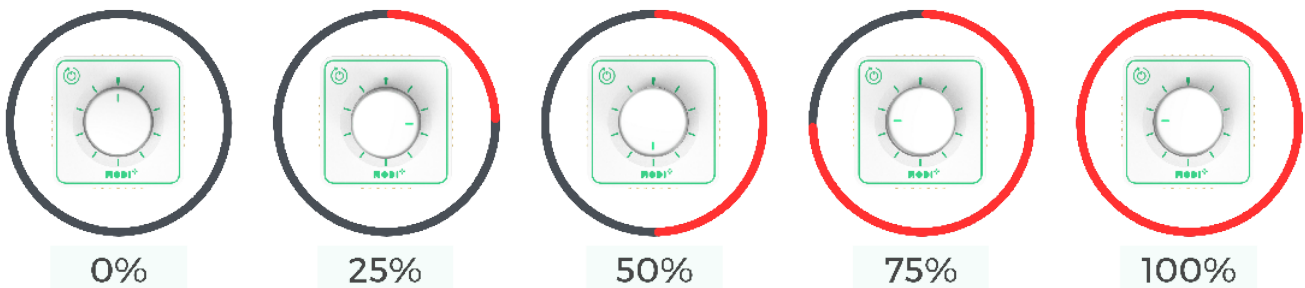
Let's look detail about the Dial module.



FUNCTIONS IN DETAIL

1. Turn (%) - Range: 0% ~ 100% This just means moving the dial around, either to the right or left.	2. Angle (°) - Range: 0~360° This is how far you've moved the dial from the default (0°), measured in degrees.
3. Mark - Range: 1~10 Marks on the dial show where it can be turned to, like the numbers on a clock.	4. Turn Speed (%) - Range: 0% ~ 100% This tells us how fast you are turning the dial, in percentage.

TURN (%)



ANGLE (°)

